

Suspension Parameter Optimization in a Paramotor UAV Using Monte Carlo Analysis

Parth Mhaske*

University of Maryland, College Park, MD 20742, USA

Christian Claudel†

University of Texas, Austin, TX 78712, USA

Paramotor Unmanned Aerial Vehicles (UAVs) are particularly suited for the autonomous transport of heavy payloads. However, when landing on unprepared runways, they can be subject to very strong impact forces that can damage them. In this article, we investigate the problem of optimizing suspension design parameters for a quad-wheeled paramotor UAV. Using a classical spring-mass-damper model for each suspension, we derive a hybrid linear system formulation of the UAV dynamics during the impact phase and use a Monte Carlo approach to optimize suspension parameters, given stochastic touchdown velocities and angles.

I. Nomenclature

k	= Spring Coefficient (N/m)
c	= Damping Coefficient (Ns/m)
L	= Spring Equilibrium Length (m)
r	= Radius of Wheels (m)
m	= Mass of UAV (kg)
l	= Length of UAV (m)
w	= Width of UAV (m)
d	= Thickness of UAV (m)
ρ	= Density of Air
C	= Drag Coefficient
A	= Surface Area of Parachute
(x, y, z)	= Coordinates of the UAV
(α, β, γ)	= Roll, pitch, yaw of the UAV
(FL, FR, BL, BR)	= Wheels

II. Introduction

Unmanned aerial vehicles (UAVs) are commonly used in mapping and surveillance applications [1]. However, to date, heavy UAVs capable of transporting large payloads are extremely expensive. These UAVs fall into two categories. Fixed-wing UAVs use a rigid wing structure to generate the required lift, while rotorcraft UAVs use rotors. A feasible alternative to fixed-wing and rotorcraft UAVs is to use a flexible wing structure, using a parafoil, an unmanned equivalent of ultralight paramotors used for recreational purposes[2].

A parafoil is a non-rigid airfoil with an aerodynamic cell structure that is inflated by the wind. The dynamic pressure associated with relative wind inflates the wing and maintains its shape into a classic wing cross-section. Parafoils have some advantages over fixed-wing and rotorcraft configurations. In particular, a parafoil weighs less than the corresponding rigid wing structure, as its shape is maintained by the dynamic pressure of the air [3]. Furthermore, compared to a fixed-wing, a parafoil is inexpensive to make and maintain. However, parafoils typically have lower lift-to-drag ratios than fixed-wing configurations, and they lack the Vertical Take Off and Landing (VTOL) capabilities of rotorcrafts [4].

*Undergraduate Student, Department of Computer Science, AIAA Student Member; pmhaske@terpmail.umd.edu.

†Associate Professor, Fariborz Maseeh Department of Civil, Architectural and Environmental Engineering; christian.claudel@utexas.edu.

In this article, we focus on the problem of simulating the landing impact of a heavy UAV equipped with a parafoil wing [5]. The UAV testbed is a “ground hopper” test article built using an ATV (all-terrain-vehicle) frame and a paraglider wing, as illustrated in Fig. 1. One of the key advantages of the paramotor configuration is the lack of wing structure, which greatly simplifies construction. In particular, the UAV itself can be a very compact structure integrating the payload, landing gear, propulsion, sensors, computers, and actuation.



Fig. 1 Paramotor ground testbed investigated in this study.

To maximize operational flexibility, we want this platform to tolerate very harsh landings that are common in highly turbulent conditions and steep approach scenarios. While parafoils are structurally light and impact-resistant (since they can deform), the paramotor structure itself can be damaged by such impacts. Energy absorption during landing impact is the central design requirement of the suspension system, following classical landing gear design principles [6].

To address this problem, we propose a hybrid linear system formulation [7] of the dynamics of the suspended mass during the impact phase. We then use this formulation to model impacts associated with various initial conditions (initial height, vertical velocity, pitch, and roll angles), and use this hybrid model formulation to robustly optimize the suspension parameters (assuming that all four suspensions are identical), under initial condition uncertainty. For the landing impact analysis presented here, the descent phase is modeled under recovery parachute only; parafoil lift during flight is outside the scope of this impact study.

The rest of the article is organized as follows. Section III contains the derivation for the system of equations that govern the behavior of the UAV during impact, Section IV presents the results of a sample simulation in order to evaluate the dynamics surrounding the impact of the UAV, Section V describes the process used to optimize the suspension parameters through a Monte Carlo analysis, and Section VI outlines the practical application of this paper.

III. Derivation

A. Assumptions

- 1) If $\alpha \sim \epsilon$ and $\beta \sim \epsilon$, then $\alpha\beta \sim \epsilon^2 \ll \epsilon$.
- 2) Mass-less Wheels
- 3) Uniform Density UAV
- 4) $\gamma = 0$
- 5) All forces are vertical
- 6) The spring-damper element is modeled as bidirectional (capable of both compression and extension forces about the equilibrium length), rather than as a unilateral ground contact. This is a standard simplification for landing-impact analysis when the UAV remains in nominal contact with the ground following touchdown.

B. Coordinate Transformation Matrices

The position of each wheel is given by:

$$\begin{pmatrix} x_i(t) \\ y_i(t) \\ z_i(t) \end{pmatrix} = \begin{pmatrix} x_{CG}(t) \\ y_{CG}(t) \\ z_{CG}(t) \end{pmatrix} + R_\alpha R_\beta \begin{pmatrix} \delta x_i \\ \delta y_i \\ 0 \end{pmatrix} \quad (1)$$

Where the rotation matrices are:

$$R_\alpha = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\alpha(t) \\ 0 & \alpha(t) & 1 \end{pmatrix} \quad (2)$$

$$R_\beta = \begin{pmatrix} 1 & 0 & \beta(t) \\ 0 & 1 & 0 \\ -\beta(t) & 0 & 1 \end{pmatrix} \quad (3)$$

After applying the small angle approximation, the z-coordinates simplify to:

$$z_{FL}(t) = z_{CG}(t) + \frac{1}{2}(\alpha(t)w - \beta(t)l) \quad (4)$$

$$z_{FR}(t) = z_{CG}(t) + \frac{1}{2}(-\alpha(t)w - \beta(t)l) \quad (5)$$

$$z_{BL}(t) = z_{CG}(t) + \frac{1}{2}(\alpha(t)w + \beta(t)l) \quad (6)$$

$$z_{BR}(t) = z_{CG}(t) + \frac{1}{2}(-\alpha(t)w + \beta(t)l) \quad (7)$$

Note that with this convention, $\beta > 0$ corresponds to a nose-down pitch attitude.

C. Spring-Damper Forces

The Mass-Spring-Damper equation for each wheel is:

$$F(t) = -kz(t) - cz'(t) \quad (8)$$

Accounting for the equilibrium length and wheel radius:

$$F_i(t) = -k(z_i(t) - L - r) - c \cdot z'_i(t) \quad (9)$$

Applying this to each wheel:

$$\begin{aligned} F_{FL}(t) = & -k \left(z_{CG}(t) + \frac{1}{2}(w\alpha(t) - l\beta(t)) - L - r \right) \\ & - c \left(\frac{1}{2}w\alpha'(t) - \frac{1}{2}l\beta'(t) + z'_{CG}(t) \right) \end{aligned} \quad (10)$$

$$\begin{aligned} F_{FR}(t) = & -k \left(z_{CG}(t) + \frac{1}{2}(-w\alpha(t) - l\beta(t)) - L - r \right) \\ & - c \left(-\frac{1}{2}w\alpha'(t) - \frac{1}{2}l\beta'(t) + z'_{CG}(t) \right) \end{aligned} \quad (11)$$

$$\begin{aligned} F_{BL}(t) = & -k \left(z_{CG}(t) + \frac{1}{2}(w\alpha(t) + l\beta(t)) - L - r \right) \\ & - c \left(\frac{1}{2}w\alpha'(t) + \frac{1}{2}l\beta'(t) + z'_{CG}(t) \right) \end{aligned} \quad (12)$$

$$F_{BR}(t) = -k \left(z_{CG}(t) + \frac{1}{2}(-w\alpha(t) + l\beta(t)) - L - r \right) - c \left(-\frac{1}{2}w\alpha'(t) + \frac{1}{2}l\beta'(t) + z'_{CG}(t) \right) \quad (13)$$

D. Equations of Motion

Applying Newton's Second Law, with aerodynamic drag opposing the direction of vertical motion:

$$\begin{aligned} F(t) &= F_{FL}(t) + F_{FR}(t) + F_{BL}(t) + F_{BR}(t) \\ &\quad - mg - \frac{1}{2}\rho |z'_{CG}| z'_{CG} (C_{UAV}A_{UAV} + C_P A_P) \\ &= mz''_{CG}(t) \end{aligned} \quad (14)$$

After simplification:

$$\begin{aligned} z''_{CG}(t) &= -\frac{4k}{m} [z_{CG}(t) - L - r] - \frac{4c}{m} [z'_{CG}(t)] \\ &\quad - g - \frac{1}{2m}\rho |z'_{CG}| z'_{CG} (C_{UAV}A_{UAV} + C_P A_P) \end{aligned} \quad (15)$$

For rotational dynamics using $\tau = r \times F = I\alpha$:

$$I_x \alpha''(t) = \frac{w}{2} (F_{FL}(t) - F_{FR}(t) + F_{BL}(t) - F_{BR}(t)) \quad (16)$$

$$I_y \beta''(t) = \frac{l}{2} (F_{BL}(t) - F_{FL}(t) + F_{BR}(t) - F_{FR}(t)) \quad (17)$$

After simplification:

$$I_x \alpha''(t) = -kw^2 \alpha(t) - cw^2 \alpha'(t) \quad (18)$$

$$I_y \beta''(t) = -kl^2 \beta(t) - cl^2 \beta'(t) \quad (19)$$

Using the moment of inertia for a uniform rectangular plate (thin-plate approximation with d^2 contributions neglected):

$$I_x = \frac{1}{12}mw^2 \quad (20)$$

$$I_y = \frac{1}{12}ml^2 \quad (21)$$

E. Final System of Equations

$$\boxed{\alpha''(t) = -\frac{12k}{m}\alpha(t) - \frac{12c}{m}\alpha'(t)} \quad (22)$$

$$\boxed{\beta''(t) = -\frac{12k}{m}\beta(t) - \frac{12c}{m}\beta'(t)} \quad (23)$$

$$\boxed{\begin{aligned} z''_{CG}(t) &= -\frac{4k}{m} [z_{CG}(t) - L - r] - \frac{4c}{m} z'_{CG}(t) - g \\ &\quad - \frac{1}{2m}\rho |z'_{CG}| z'_{CG} (C_{UAV}A_{UAV} + C_P A_P) \end{aligned}} \quad (24)$$

F. State-Space Formulation

We define the state variables as:

$$\alpha_1(t) = \alpha(t), \quad \alpha_2(t) = \alpha'(t) \quad (25)$$

$$\beta_1(t) = \beta(t), \quad \beta_2(t) = \beta'(t) \quad (26)$$

$$z_1(t) = z_{CG}(t), \quad z_2(t) = z'_{CG}(t) \quad (27)$$

The state-space equations become:

$$\alpha'_1(t) = \alpha_2(t) \quad (28)$$

$$\alpha'_2(t) = -\frac{12k}{m}\alpha_1(t) - \frac{12c}{m}\alpha_2(t) \quad (29)$$

$$\beta'_1(t) = \beta_2(t) \quad (30)$$

$$\beta'_2(t) = -\frac{12k}{m}\beta_1(t) - \frac{12c}{m}\beta_2(t) \quad (31)$$

$$z'_1(t) = z_2(t) \quad (32)$$

$$z'_2(t) = -\frac{4k}{m}[z_1(t) - L - r] - \frac{4c}{m}z_2(t) - g - \frac{1}{2m}\rho|z_2(t)|z_2(t)(C_{UAV}A_{UAV} + C_P A_P) \quad (33)$$

These equations represent the all-wheels-active mode of the hybrid system; the simulation evaluates analogous mode-specific equations for each of the 16 wheel-contact configurations as individual wheels engage and disengage during the impact phase.

IV. Testing

In this section, we present the result of a Python simulation that was run to analyze the landing of the UAV. The purpose of this simulation is to evaluate the dynamic behavior of the UAV during its landing. The assumptions that were taken during the derivation section will still apply to this section.

This simulation was created using the ODEINT function from the SciPy library [8], which provides standard numerical integration of ordinary differential equations [9]. The functions derived in the previous section were used in order to model the forces, velocity, and position of the UAV. In order to make sure that some of the values in the simulation would not be affected by free fall, switches in the code were implemented so that those values would only start to get calculated after that specific wheel hit the ground. We refer to this as the wheel's "activation time."

We examine a scenario in which the UAV has a mass of 140 kg and is released from a height of 1.9 meters beneath a recovery parachute. The parachute is modeled as a pure drag element with drag coefficient $C_P = 0.75$ (typical of a flat circular canopy) and reference area $A_P = 70 \text{ m}^2$, producing a terminal velocity of approximately 6.5 m/s at the UAV's mass. This drag model represents the descent configuration only; no parafoil lift is included, consistent with a post-flare or recovery-chute scenario. The spring has a constant of 8000 N/m and the damper will have a constant of 212 Ns/m, which gives a damping ratio ($\zeta = c/\sqrt{km}$) of approximately 0.20 for the vertical mode. The free length for the springs will be 0.50 meters, with a solid length of 0.20 meters. Additionally, the UAV will have a length of 1.50 meters, a width of 0.60 meters, and a wheel radius of 0.20 meters. The mass, spring free/solid length, length, width, and wheel radius will remain constant for the rest of the sections.

The array of initial conditions will be: $[\alpha, \alpha', \beta, \beta', z_{CG}, z'_{CG}]$. For this particular simulation, the array was: $[0, 0.1, 0, -0.1, 2.6, 0]$. The value z_{CG} is the sum of L , r , and the initial height that the UAV is being dropped from. This allows us to simulate the fall of the UAV while also taking into consideration the extra height that the spring and wheels would be adding.

With the initial conditions and parameters set, we proceed to run the simulation to observe the behavior of the UAV and determine whether the simulation would be considered successful. We define a successful simulation as one in which the maximum force on any of the springs does not exceed $\frac{mg}{4} + k \cdot (0.30 \text{ m})$, corresponding to the static load per wheel plus the elastic spring force at maximum allowable compression (0.30 m of travel between free and solid length), and the minimum length of any of the springs does not fall below 0.20 meters. The results from the simulation were plotted using Matplotlib [10]. In the following graphs, we analyze the key variables and switches to assess the UAV's collision with the ground.

We first start with the graph of the position and velocity of $\alpha(t)$ and $\beta(t)$:

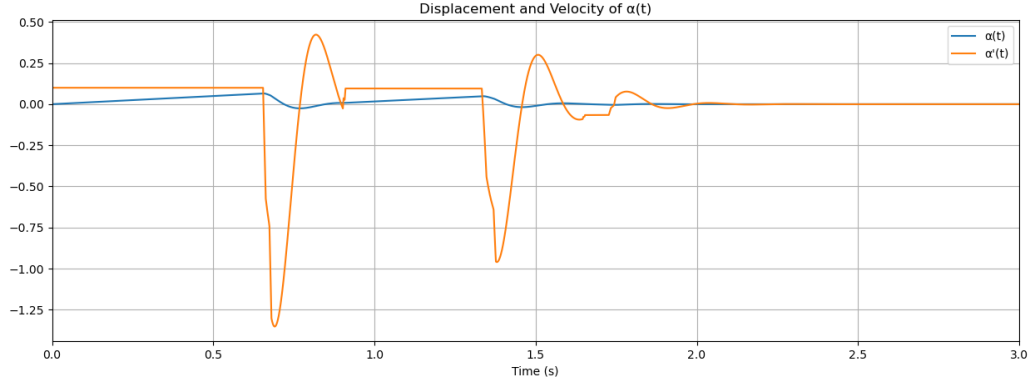


Fig. 2 Roll angle (α) position and velocity over time.

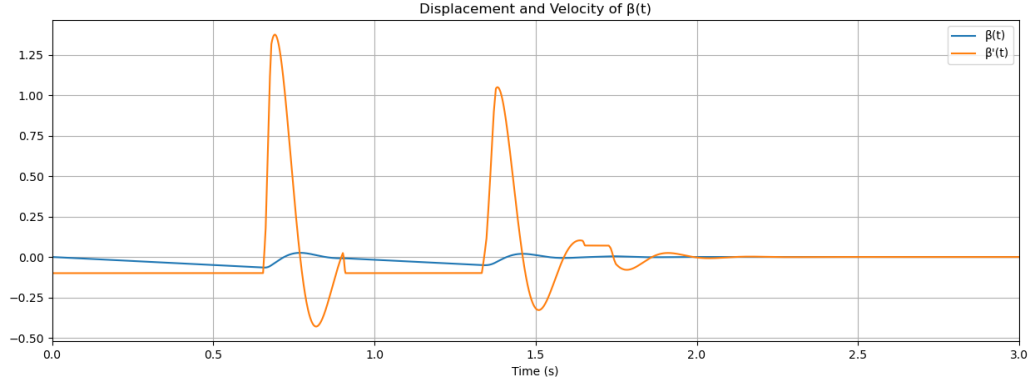


Fig. 3 Pitch angle (β) position and velocity over time.

Since the absolute values of $\alpha'(0)$ and $\beta'(0)$ are the same, the shapes of the graphs are similar. During the initial free fall, $\alpha(t)$ shows a constant increase while $\beta(t)$ shows a constant decrease. After the UAV initially hits the ground, α and β experience an acceleration, and reach their respective maximum values for both velocity and displacement of around ± 1.36 rad/s and ± 0.07 radians. The initial acceleration lasts for around 0.21 seconds before the velocities for both α and β both go back to 0.1 rad/s and -0.1 rad/s respectively. This is because of the under-damped nature of this suspension system, which causes the UAV to “bounce back” upwards. This cycle of acceleration and constant velocity continues two more times, as shown.

This oscillatory behavior not only affects the rotational dynamics of the UAV but also the vertical velocity and position of the springs.

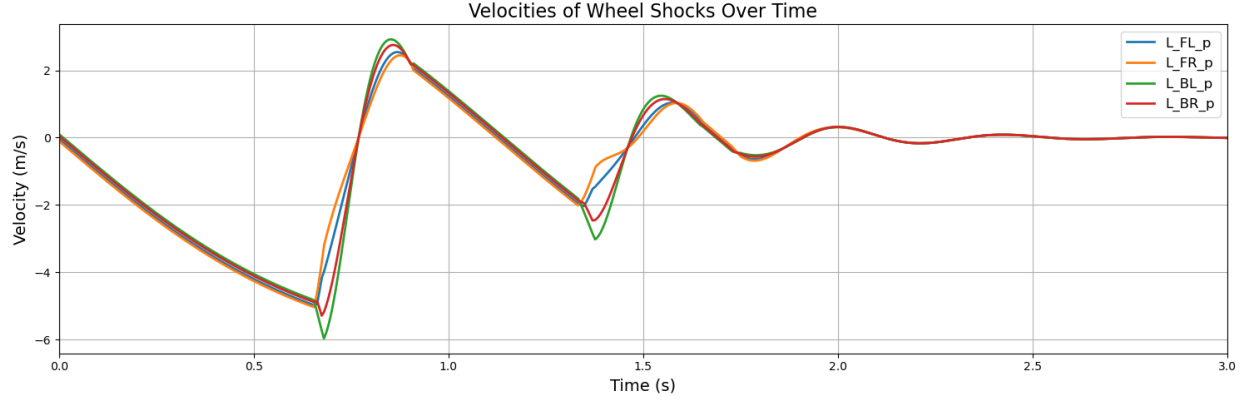


Fig. 4 Spring velocity profiles during landing.

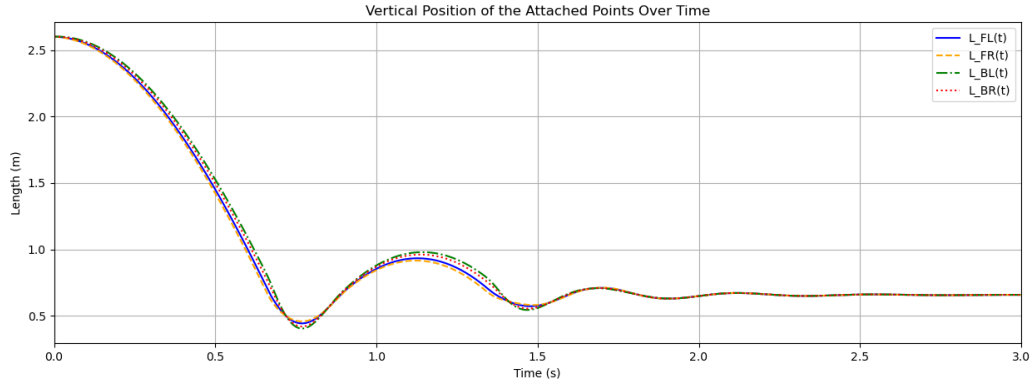


Fig. 5 Vertical position of attachment points with wheel activation times.

In Fig. 5, we see the vertical position of the attached points and how that changes over time. Attached points are defined as the point where the spring attaches to the body of the UAV. Therefore, even though the UAV is dropped from a height of 1.9 meters, the height of the attachment point is 2.6m. Due to the drag force from the parachute during initial free fall, we see in Fig. 4 that the velocity is not linear, and this is reflected in Fig. 5. When the wheels first touch the ground at the first activation time, the springs get compressed and reach a minimum value ranging from 0.21-0.26m. This range is due to the rotations caused by the nonzero values of the roll and pitch. After the first activation point, the reaction forces cause the UAV to bounce upwards twice. Though, each subsequent time, the reaction force decreases in value. Additionally, the velocity when the UAV is falling back down is closer to linear, as drag would play less of a role at lower speeds. At the end, we see that the final length of the spring is 0.46 meters. This matches the static spring length predicted by Hooke's law, $L - (mg/4)/k$, since the four springs equally distribute the UAV weight at rest.

Finally, we analyze the forces in order to determine whether or not this simulation can be labelled as successful.



Fig. 6 Forces on each spring during landing impact.

The graph in Fig. 6 begins at the first activation time for each spring/shock. A positive force is defined as a compressive force and a negative force is defined as when the spring is stretched. The dotted vertical lines represent the activation times for each of the springs and also correspond with the force discontinuities as each wheel engages. When examining the forces, the differences due to the roll and pitch are more pronounced as these rotations caused an uneven force distribution between the four springs at the start. Though, as the wheels eventually settle after the final activation point, the differences between the springs are less defined and eventually become zero.

The minimum spring length observed was 0.204 m, above the solid length of 0.20 m, and the maximum force observed was 2585.93 N, which remained below the failure threshold of $\frac{mg}{4} + k \cdot (0.30 \text{ m}) = 2743.35 \text{ N}$. This simulation was therefore successful.

Table 1 on the following page summarizes the key collected values from this test.

Table 1 Min and Max Values for Various Parameters

Parameter	Min Value	Max Value
$\alpha(t)$	-0.02533	0.06553
$\beta(t)$	-0.06562	0.02576
$z_{CG}(t)$	0.43124	2.60000
$\alpha'(t)$	-1.35237	0.42301
$\beta'(t)$	-0.43008	1.37498
$z'_{CG}(t)$	-4.92845	2.63808
Final Displacement $z_{CG}(t)$	0.65708	
Final Velocity $z'_{CG}(t)$	3.53×10^{-5}	
Spring Lengths (L)	Min Value	Max Value
$L_{FL}(t)$	0.24296	2.40000
$L_{FR}(t)$	0.25816	2.40000
$L_{BL}(t)$	0.20433	2.40056
$L_{BR}(t)$	0.21953	2.40010
Forces (N)	Min Value	Max Value
Force at L_{FL} Activation	317.38	
Force at L_{FR} Activation	236.93	
Force at L_{BL} Activation	1465.04	
Force at L_{BR} Activation	1490.55	
F_{FL} after activation	-2260.56	2585.93
F_{FR} after activation	-2128.27	2507.77
F_{BL} after activation	-1891.31	2184.30
F_{BR} after activation	-1753.85	2026.46
Final Force for all wheels	343.34	

V. Discussion

However, there are instances in which the current configuration of this spring-mass-damper system would result in a failed simulation. For example, if the height that the UAV was dropped was increased to 2 meters, the simulation would fail since the back left spring would compress to a length of 0.19 meters, causing the spring to break as it has been compressed past its solid length. In order to rectify this, we sought out to find a spring and damper constant that would perform well across the sampled touchdown configurations. In order to do that, a Monte Carlo Simulation [11, 12] was created in Python using NumPy [13] and SciPy [8].

A. Monte Carlo Simulation Process

To optimize the suspension parameters, we used a Monte Carlo simulation approach. The process involves:

- 1) Generating random spring parameters (k, c) with $k \in [5000, 30000]$ N/m and $c \in [100, 1500]$ Ns/m.
- 2) Running simulations with randomized touchdown velocity, roll/pitch angles, and angular rates.
- 3) Checking if the roll (α) or pitch (β) exceed predefined bounds:
 - If out of bounds, the trial is recorded as a failure.
- 4) Calculating activation times, forces, and spring lengths.
- 5) Recording failures or successes based on the following criteria:
 - Failure: Spring compression below solid length, force exceeding structural limits, roll or pitch exceeding ± 0.2 rad, or vertical velocity at $t = 2$ s exceeding 0.3 m/s.
 - Success: Safe landing meeting all of the above criteria.
- 6) Repeating for 5000 parameter combinations, with 1000 trials each.

The optimal parameters were identified as those yielding the highest success count, with ties broken by the lowest average maximum force. Figure 7 shows the results of this optimization process.

Note that the Monte Carlo samples the touchdown velocity directly rather than a drop height. The sample simulation in Section IV used a 1.9 m drop because that is the test article's bench-test scenario, but operationally the UAV descends under a deployed recovery parachute. Once the chute reaches steady state the descent rate is fixed at the parachute's terminal velocity (approximately 6.5 m/s for the modeled chute), regardless of deployment altitude. A drop from 50 meters or 500 meters produces the same impact velocity as a drop from any altitude that is sufficient for the chute to stabilize. The sampled range $v_{td} \in [4.0, 7.5]$ m/s accounts for the operational variation around terminal: the lower bound covers pre-impact flare maneuvers and updrafts that bleed off velocity, and the upper bound covers downdrafts and partial-canopy deployment that increase ground-frame impact velocity beyond steady-state terminal.

The optimal combination of parameters was when $k = 21,216$ N/m and $c = 484.56$ Ns/m. This specific combination had a success count of 1000 out of 1000, demonstrating that this system will be able to perform reliably under these conditions. The average minimum spring length for this configuration was 0.310 meters. Additionally, the average maximum force experienced by the system was 5057 N, which remained below the failure threshold of $\frac{mg}{4} + k \cdot 0.30 \text{ m} = 6708$ N, further supporting this combination's reliability under the sampled conditions. While other configurations within the high-success region (for example, k approaching 30,000 N/m at moderate damping) also achieved 1000/1000 success, peak impact force scales with spring stiffness, so the lowest-force perfect cell sits at the lower-stiffness end of the perfect-success region. The selection therefore identifies the configuration that meets the success criterion with the largest structural margin, rather than simply the highest success count.

B. Analysis of Results

The optimization revealed several important trends:

- 1) *Low Spring Constants*: Configurations with $k < 14,000$ N/m showed poor performance due to excessive compression and bottoming out of the suspension at high-velocity touchdowns.
- 2) *High Damping Coefficients*: Systems with damping coefficients above approximately 1000 Ns/m experienced higher peak forces, because excessive damping causes the suspension to behave more rigidly and transmit impact loads directly to the structure rather than absorbing them through controlled oscillation.
- 3) *Low Damping Coefficients*: Systems with damping coefficients below approximately 250 Ns/m failed to dissipate post-impact energy within the simulation window, producing repeated bounce cycles that drove the roll and pitch angles outside operational bounds.
- 4) *Optimal Region*: The best performance occurred with moderate-to-high spring constants (18000–30000 N/m) and moderate damping coefficients (400–840 Ns/m), providing a balance between force limitation and displacement control.

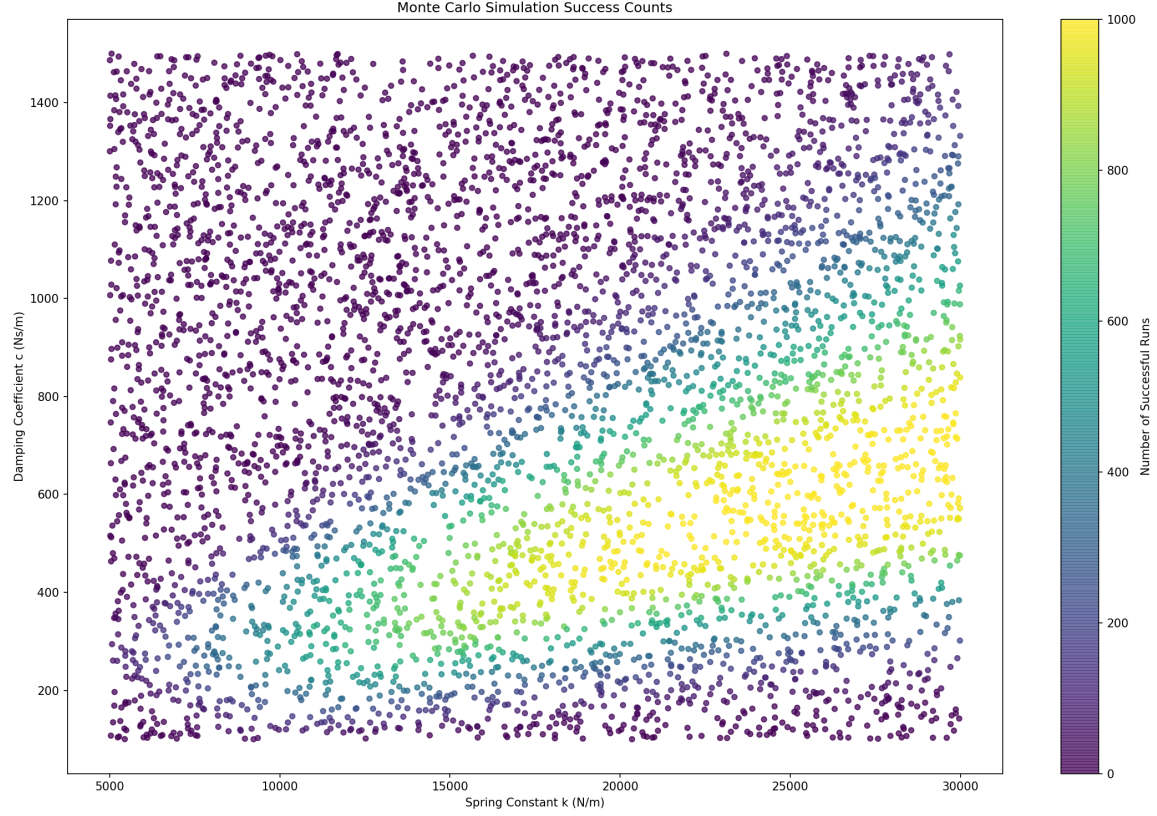


Fig. 7 Monte Carlo optimization results for suspension parameters.

The optimal damping ratio of 0.28 is consistent with that of typical vehicle suspension systems [14, 15], which is appropriate given the more severe impact conditions experienced by the UAV.

VI. Conclusion

In this article, we analyzed the impact resistance of a heavy paramotor UAV during the landing phase. We formulated the dynamics of the paramotor UAV with a rigid body frame model, with independent suspensions that share identical parameters. Since pitch and roll angles are assumed to be relatively low on impact, the resulting model is linear for each particular configuration (set of wheels that have impacted the ground), with the exception of the quadratic aerodynamic drag term in the vertical equation. Since the configuration changes discretely over time during the impact phase, the resulting model is hybrid and exhibits 16 possible modes that correspond to all possible wheel contact configurations. Since this model is not amenable to classical linear system analysis, we focused on a Monte Carlo approach, in which parameters are to minimize the probability of UAV damage, which is encoded by upper limits in reaction forces for each attach point (structural limits) and lower limits in the position of any point of the frame during the impact phase (no-ground contact condition).

This Monte Carlo simulation allows us to find the optimal elastic and damping parameters of the suspension for a given geometry of the UAV frame. These results will be implemented in a heavy paramotor-based UAV currently under development at the University of Texas, Austin.

References

- [1] Nex, F., and Remondino, F., “UAV for 3D mapping applications: a review,” *Applied Geomatics*, Vol. 6, 2014, pp. 1–15. <https://doi.org/10.1007/s12518-013-0120-x>.
- [2] Stein, J., Madsen, C., and Strahan, A., “An Overview of the Guided Parafoil System Derived from X-38 Experience,” *18th AIAA Aerodynamic Decelerator Systems Technology Conference and Seminar*, Munich, Germany, 2005. <https://doi.org/10.2514/6.2005-1652>.
- [3] Slegers, N., and Costello, M., “Aspects of Control for a Parafoil and Payload System,” *Journal of Guidance, Control, and Dynamics*, Vol. 26, No. 6, 2003. <https://doi.org/10.2514/2.6933>.
- [4] Carruthers, A. C., and Filippone, A., “Aerodynamic Drag of Parafoils,” *Journal of Aircraft*, Vol. 42, No. 4, 2005. <https://doi.org/10.2514/1.12255>.
- [5] Wachlin, J., and Costello, M., “Simulation of the Landing Dynamics of a Guided Airdrop System,” *AIAA Aviation 2019 Forum*, 2019. <https://doi.org/10.2514/6.2019-3372>.
- [6] Currey, N. S., *Aircraft Landing Gear Design: Principles and Practices*, AIAA Education Series, American Institute of Aeronautics and Astronautics, Washington, DC, 1988. <https://doi.org/10.2514/4.861468>.
- [7] Goebel, R., Sanfelice, R. G., and Teel, A. R., “Hybrid Dynamical Systems,” *IEEE Control Systems Magazine*, Vol. 29, No. 2, 2009, pp. 28–93. <https://doi.org/10.1109/MCS.2008.931718>.
- [8] Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., Carey, C. J., Polat, İ., Feng, Y., Moore, E. W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E. A., Harris, C. R., Archibald, A. M., Ribeiro, A. H., Pedregosa, F., van Mulbregt, P., and SciPy 1.0 Contributors, “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods*, Vol. 17, 2020, pp. 261–272. <https://doi.org/10.1038/s41592-019-0686-2>.
- [9] Press, W. H., Teukolsky, S. A., Vetterling, W. T., and Flannery, B. P., *Numerical Recipes: The Art of Scientific Computing*, 3rd ed., Cambridge University Press, New York, 2007.
- [10] Hunter, J. D., “Matplotlib: A 2D graphics environment,” *Computing in Science & Engineering*, Vol. 9, No. 3, 2007, pp. 90–95. <https://doi.org/10.1109/MCSE.2007.55>.
- [11] Metropolis, N., and Ulam, S., “The Monte Carlo Method,” *Journal of the American Statistical Association*, Vol. 44, 1949, pp. 335–341. <https://doi.org/10.1080/01621459.1949.10483310>.
- [12] Rubinstein, R. Y., and Kroese, D. P., *Simulation and the Monte Carlo Method*, 3rd ed., John Wiley & Sons, 2016. <https://doi.org/10.1002/9781118631980>.
- [13] Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., and Oliphant, T. E., “Array programming with NumPy,” *Nature*, Vol. 585, No. 7825, 2020, pp. 357–362. <https://doi.org/10.1038/s41586-020-2649-2>, URL <https://doi.org/10.1038/s41586-020-2649-2>.
- [14] Calvo, J., Diaz, V., and San Román, J., “Establishing inspection criteria to verify the dynamic behaviour of the vehicle suspension system by a platform vibrating test bench,” *International Journal of Vehicle Design*, Vol. 38, 2005. <https://doi.org/10.1504/IJVD.2005.007623>.
- [15] Cole, D. J., “Fundamental Issues in Suspension Design for Heavy Road Vehicles,” *Vehicle System Dynamics*, Vol. 35, No. 4-5, 2001, pp. 319–360. <https://doi.org/10.1076/vesd.35.4.319.2045>.